MIL-M-28787/3B(EC)
3 June 1982
SUPERSEDING
MIL-M-28787/3A(EC)
1 August 1979

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE GDA

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389 except as modified in 3.4.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3 REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 2.06 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389 except the operating temperature range shall be 0° to 80°C..

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be GDA.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3 5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3 5.8 Marking Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/3-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3 6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4 3 Test equipment Test equipment specified on figures and in tables with a type designator shall be as specified in MI -STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4 1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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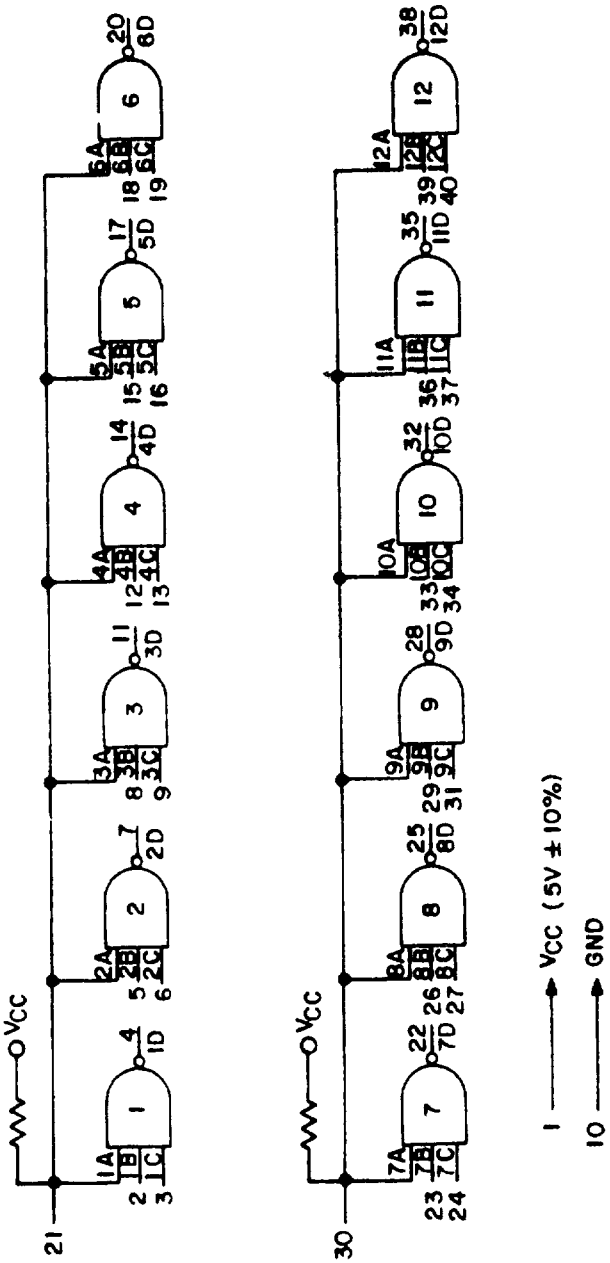


FIGURE 1. GDA module function diagram, NAND gate.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 80°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A12	V_{OUT} (low loaded)	III	S	0.1	0.8	0.1	0.8	0.1	0.8	0.1	0.8	V
B1-B24	Input current	IV	S	0.0	0 5	0.0	0.5	0.0	0.5	0.0	0.5	mA
B25,B26	Input current	IV	S	0	3	0	3	0	3	0	3	mA
C1,C2	Strobe resistor	V	S	2,160	2,640	2,160	2,640	2,160	2,640	2,160	2,640	Ω
D1-D24	Input clamp diode voltage	VI	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D25,D26	nput clamp diode voltage	VI	S	-0.4	-1.5	-0.4	-1.51	-0.4	-1.5	-0.4	-1.5	V
E1-E12	Output voltage, low level	VII	X	0	450	0	450	0	450	0	450	mV
F1-F36	Output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
G1-G24	Input current, low level	VIII	X	1.0	2 2	1.0	2.2	1.0	2.2	1.0	2.2	mA
G25,G26	Input current, low level	VIII	X	6.0	15.4	6.0	15.4	6.0	15.4	6.0	15.4	mA
H1-H48	Input current, high level	VIII	X	0	60	0	60	0	60	0	60	µA
H49,H50	Input current, high level	VIII	X	-0.80	-1.15	-0.80	-1.15	-0.80	-1.15	-0.80	-1.15	mA
J1-J12	Output short circuit current	VIII	X	40	100	35	100	35	100	25	100	mA
K1	Power supply current, low level	IX	S	30	120	30	120	30	120	30	120	mA
K2	Power supply current, high level	IX	S	16	52	16	52	16	52	16	52	mA
L1-L12	Transition time (t_{THL})	X	S	3	20	3	20	3	25	3	25	ns
M1-M12	Transition time (t_{TLH})	XI	S	3	20	3	20	3	25	3	25	ns
N1-N12	Propagation delay time (t_{PLH})	XII	S	3	20	3	20	3	25	3	25	ns

See footnotes at end of table

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 80°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P1-P12	Propagation delay time (t_{PHL})	XIII	S	3	20	3	20	3	25	3	25	ns
Q	Power supply transient amplitude	XIV	Q	0	400	0	400	0	400	0	400	mV
R	Isolation	XV	S	10	∞	10	∞	10	∞	10	∞	MΩ
S1-S12	Output crosstalk, low level	XVI	Q	0	400	0	400	0	400	0	400	mV
T1-T12	Output crosstalk, high level	XVI	Q	0	400	0	400	0	400	0	400	mV
U1-U24	Input capacitance	XVII	Q	2	12	2	12	2	12	2	12	pF
U25,U26	Input capacitance	XVII	Q	15	60	15	60	15	60	15	60	pF
V	Power supply transient rise time	XVIII	S	2.5	7.5	2.5	7.5	2.5	7.5	2.5	7.5	µs

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows. A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through V in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through P, S, T, and V shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through V shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through V shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military Specification MIL-M-28787/3 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658753. Part number M28787/3-1 supersedes part numbers 2658753 and M28787/3.

* 6.4 Module description. The following electrical parameters reflect the 0° to 80°C end-of life limits from table I.

Supply voltage	5.0 Vdc ±10%
Maximum power dissipation	
All output low level	660 mW (outputs not loaded)
All outputs high level	286 mW (outputs not loaded)
Speed	DC to 20 MHz

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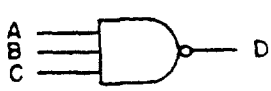
Input threshold voltage	
Low level	0.8 V maximum
High level	2.0 V minimum
Input current	
Low level (V_{IL} = 0 V)	2.2 mA maximum
High level (V_{IH} = 2.4 V)	60 µA maximum
Input current, strobe	
Low level (V_{IL} = 0 V)	6.0 mA minimum, 15.4 mA maximum.
High level (V_{IH} = 2.4 V)	-0.8 mA minimum, -1.5 mA maximum.
Output voltage	
Low level	0.45 V maximum
High level	2.4 V minimum
Output drive current	
Low level	22 mA maximum
High level	0.6 mA maximum

* 6.5 Module function. This module contains two groups of six strobable two-input NAND gates. Each group shall have one common strobe input.

6.5.1 Input to output relationship. Each gate has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship

Input	Output
Any inputs at low level All inputs at high level	High level Low level



$D = \overline{ABC} = \bar{A} + \bar{B} + \bar{C}$

6 5.2 Parallel outputs. Paralleling of outputs is not permissible.

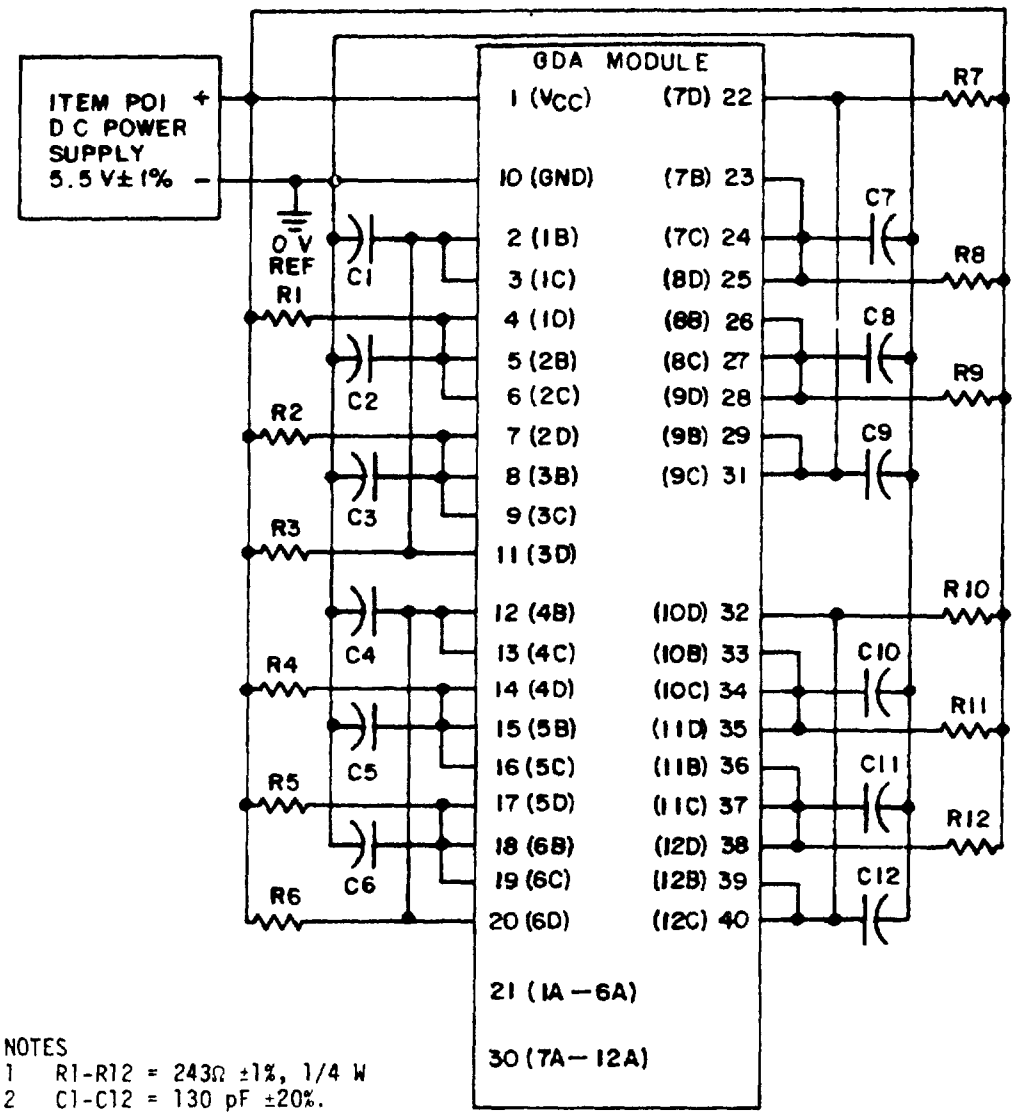
* 6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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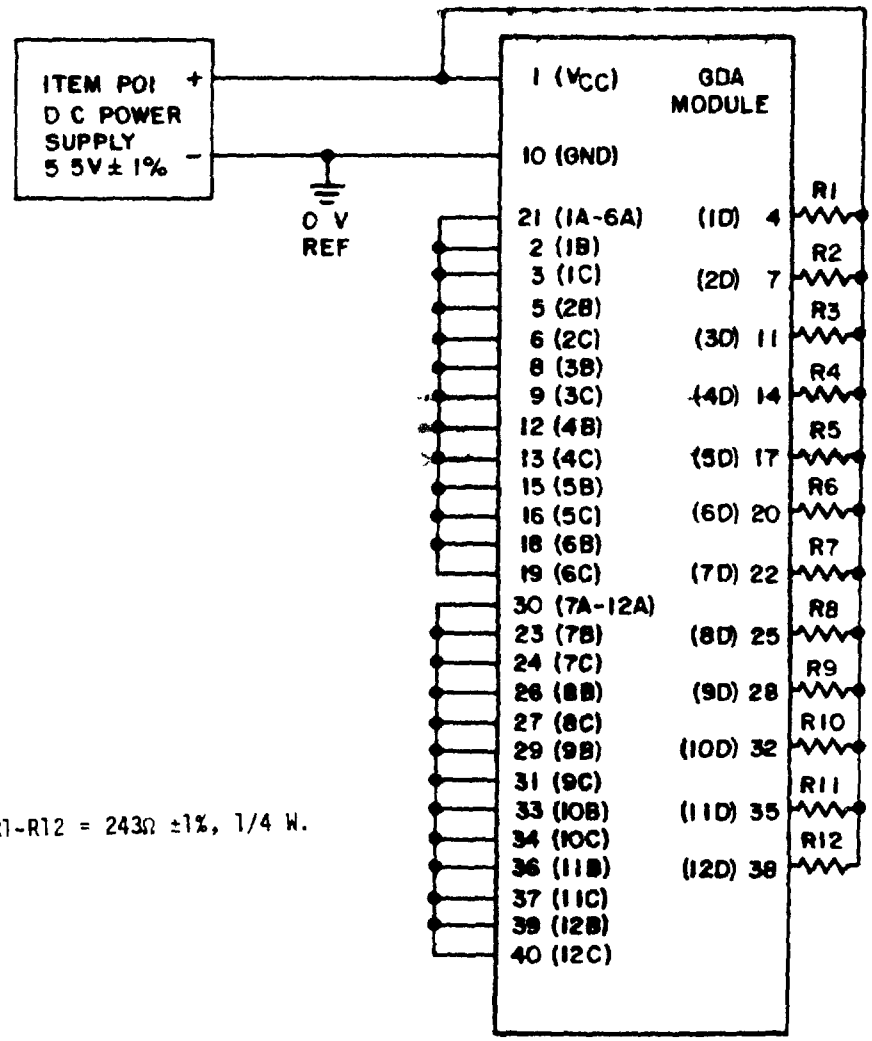
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NOTES
1 R1-R12 = 243Ω ±1%, 1/4 W
2 C1-C12 = 130 pF ±20%.

FIGURE 2. Operating temperature and life test circuit.

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NOTE R1-R12 = 243Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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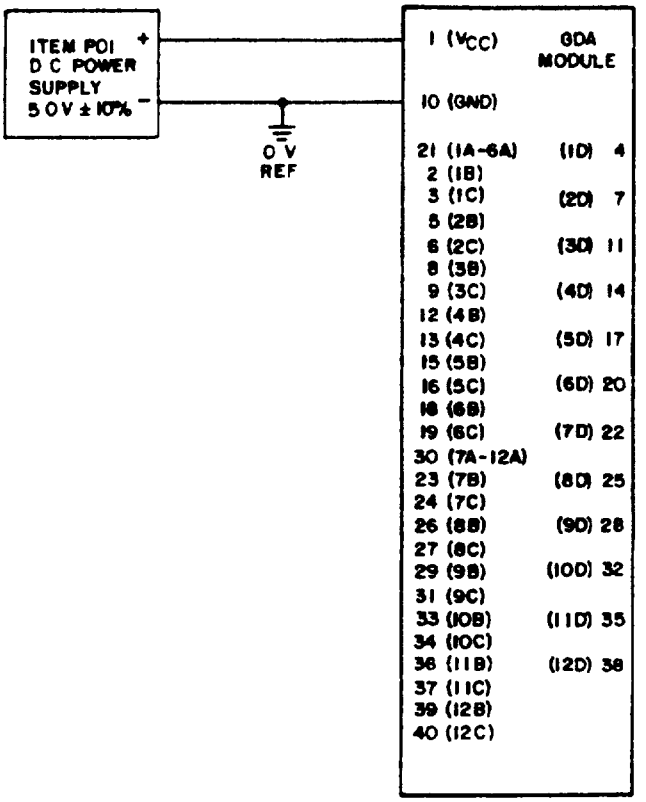


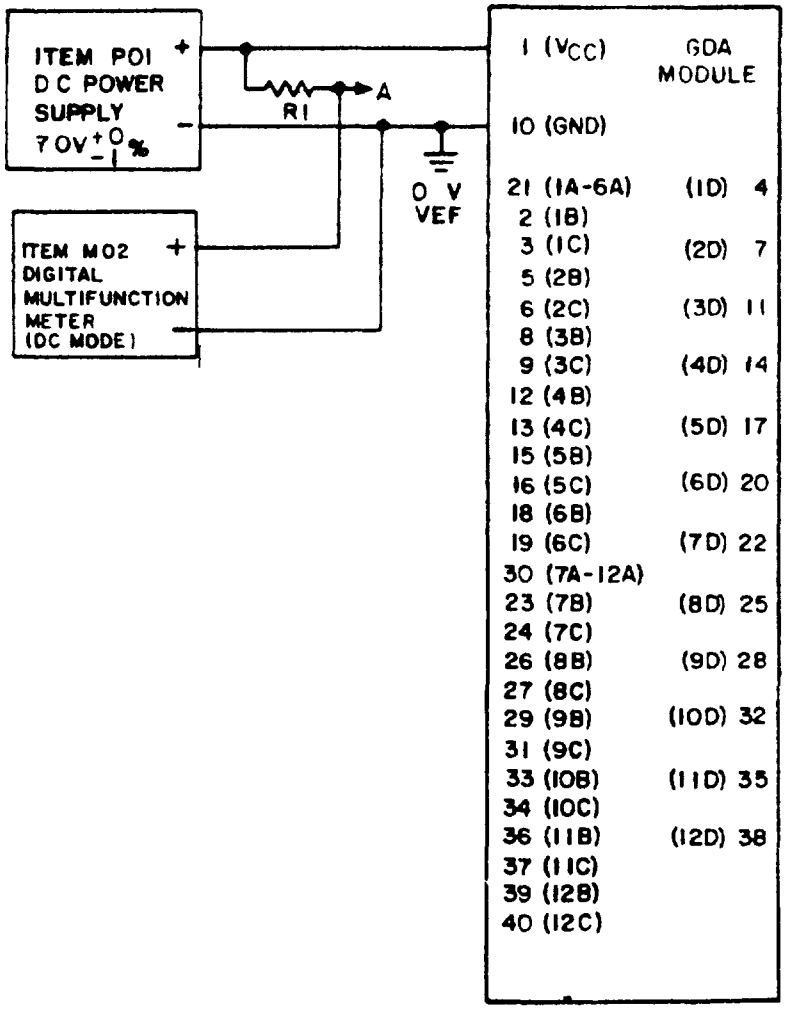
FIGURE 4 Vibration test circuit

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
A1	1D	V_{OUT} (low loaded)	4	0.1	0.8	V	5
A2	2D	↓	7	↓	↓	↓	↓
A3	3D		11				
A4	4D		14				
A5	5D		17				
A6	6D		20				
A7	7D		22				
A8	8D		25				
A9	9D		28				
A10	10D		32				
A11	11D		35				
A12	12D		38				

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NOTES

* 1. R1 = 243Ω ±1%, 1/4 W.

2. After point A has been connected to the appropriate module pin, the output voltage shall be monitored for a minimum of five seconds before the measurement is released.

FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input current test.

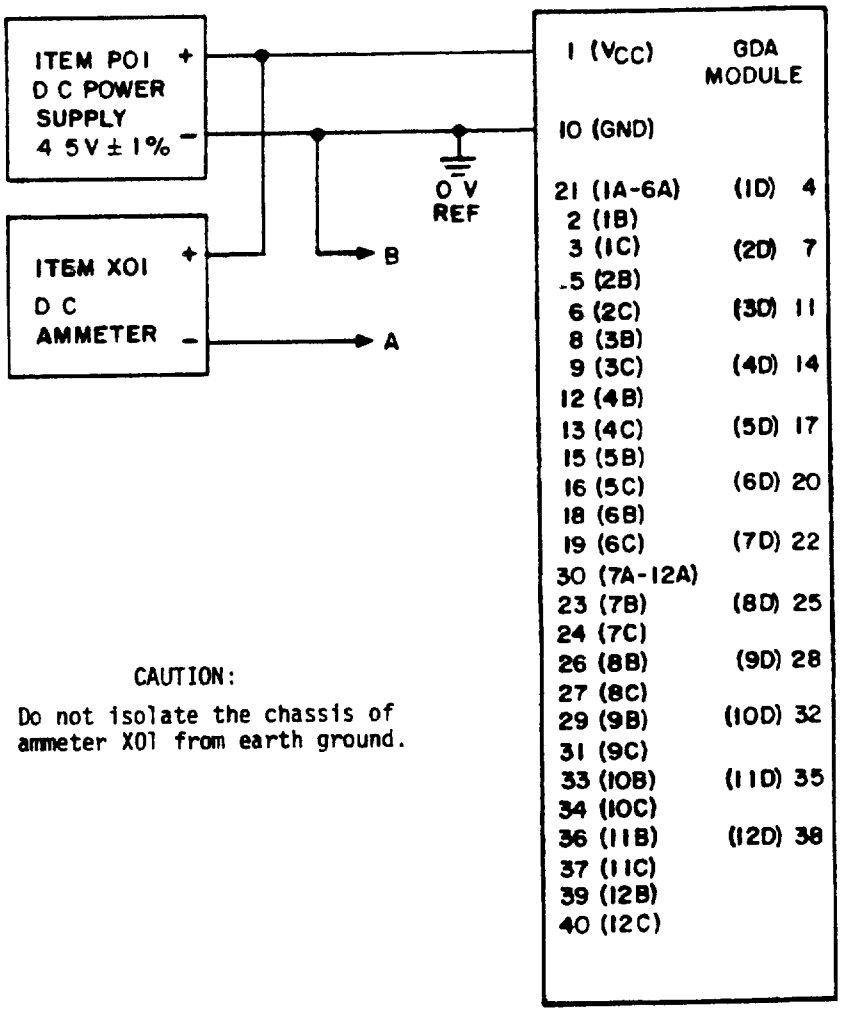
Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B1	1B	Input current	2	3, 21	0.0	0.5	mA	6
B2	1C		3	2, 21				
B3	2B		5	6, 21				
B4	2C		6	5, 21				
B5	3B		8	9, 21				
B6	3C		9	8, 21				
B7	4B		12	13, 21				
B8	4C		13	12, 21				
B9	5B		15	16, 21				
B10	5C		16	15, 21				
B11	6B		18	19, 21				
B12	6C		19	18, 21				
B13	7B		23	24, 30				
B14	7C		24	23, 30				
B15	8B		26	27, 30				
B16	8C		27	26, 30				

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TABLE IV Input current test - Continued.

Test condition	Parameter		Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
B17	9B	Input current	29	31, 30	0 0	0 5	mA	6
B18	9C		31	29, 30				
B19	10B		33	34, 30				
B20	10C		34	33, 30				
B21	11B		36	37, 30				
B22	11C		37	36, 30				
B23	12B		39	40, 30				
B24	12C		40	39, 30				
B25	1-6A		21	2, 3, 5, 6, 8, 9, 12, 13, 15, 16, 18, 19	0	3		
B26	7-12A		30	23, 24, 26, 27, 29, 31, 33, 34, 36, 37, 39, 40	0	3		

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

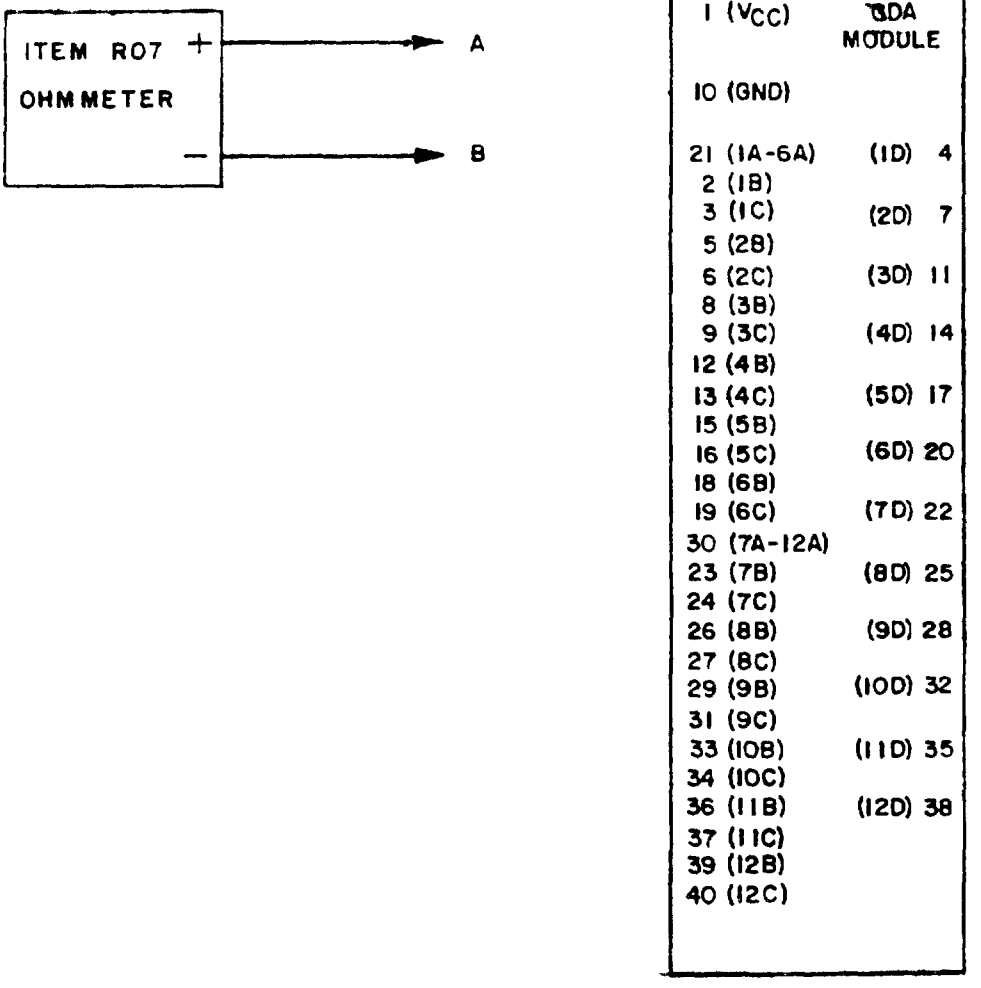
FIGURE 6. Input current test circuit for table IV.

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* TABLE V. Strobe resistor test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Strobe resistor	21	1	2,160	2,640	Ω	7
C2	Strobe resistor	30	1	2,160	2,640	Ω	7

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* FIGURE 7. Strobe resistor test circuit for table V.

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TABLE VI Input clamp diode voltage test.

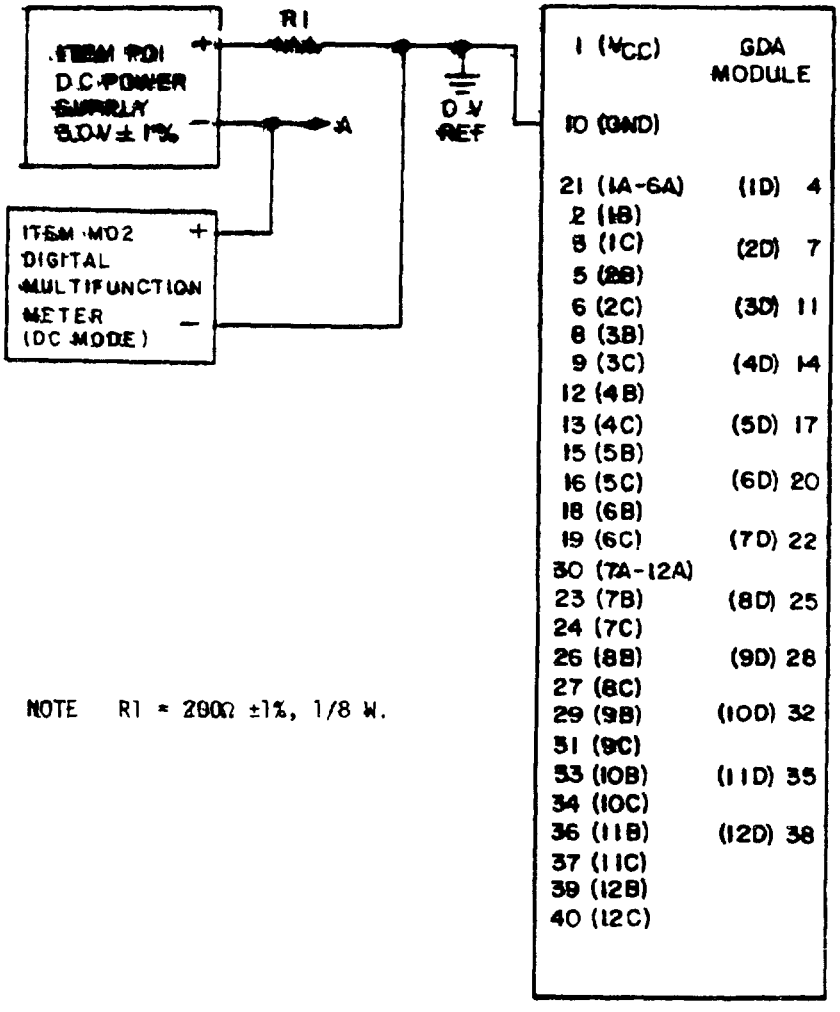
Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
D1	1B	Input clamp diode voltage	2	-0.4	-1.5	V	8
D2	1C		3				
D3	2B		5				
D4	2C		6				
D5	3B		8				
D6	3C		9				
D7	4B		12				
D8	4C		13				
D9	5B		15				
D10	5C		16				
D11	6B		18				
D12	6C		19				
D13	7B		23				
D14	7C		24				
D15	8B		26				
D16	8C		27				
D17	9B		29				

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TABLE VI. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect A to pin	Initial limits 25 ±5°C ambient Min	Initial limits 25 ±5°C ambient Max	Unit	Test figure
D18	9C Input clamp diode voltage	31	-0.4	-1.5	V	8
D19	10B	33				
D20	10C	34				
D21	11B	36				
D22	11C	37				
D23	12B	39				
D24	12C	40				
D25	1-6A	21				
D26	7-12A	30				

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NOTE R1 = 200Ω ±1%, 1/8 W.

FIGURE 8 Input clamp diode voltage test circuit for table VI

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* TABLE VII. Output voltage (low-level and high-level) test.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
E1	1D	Output voltage, low level	2, 3, 21	4	0	450	mV	9
E2	2D	↓	5, 6, 21	7	↓	↓	↓	↓
E3	3D		8, 9, 21	11				
E4	4D		12, 13, 21	14				
E5	5D		15, 16, 21	17				
E6	6D		18, 19, 21	20				
E7	7D		23, 24, 30	22				
E8	8D		26, 27, 30	25				
E9	9D		29, 31, 30	28				
E10	10D		33, 34, 30	32				
E11	11D		36, 37, 30	35				
E12	12D		39, 40, 30	38				

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TABLE VII Output voltage (low-level and high-level) test - Continued.

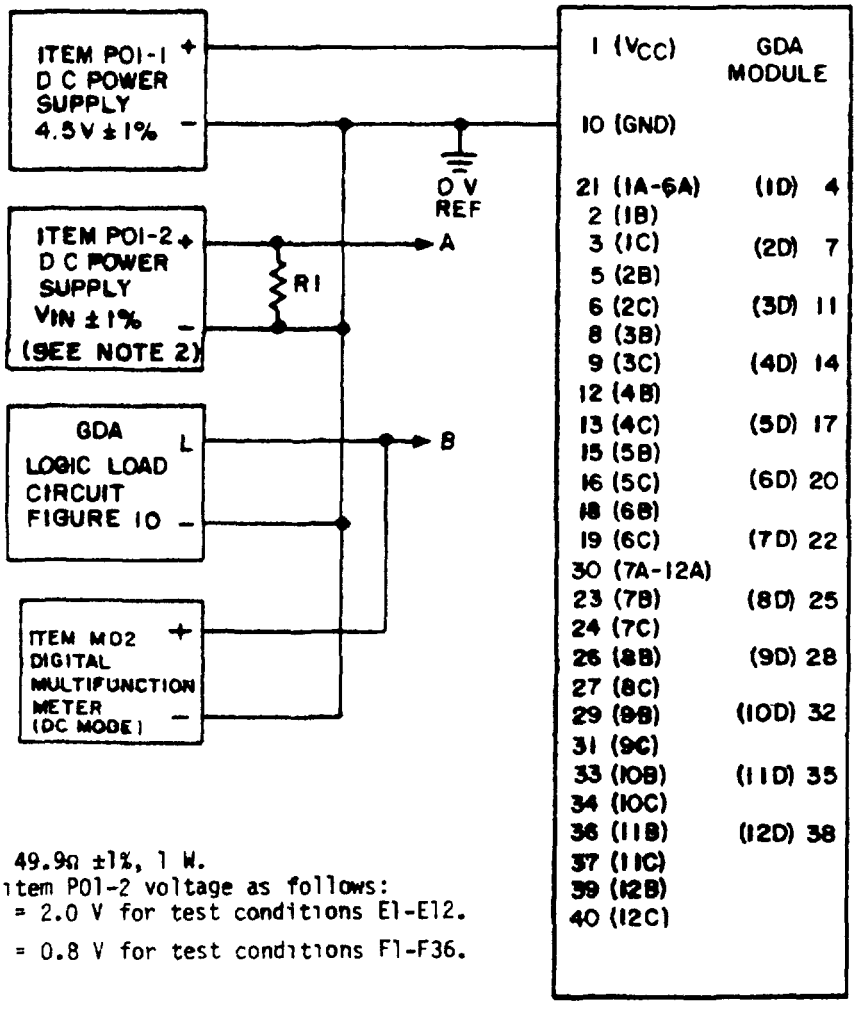
Test condition		Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
F1	1D	Output voltage, high level	21	4	2.4	4.5	V	9
F2	1D		2	4				
F3	1D		3	4				
F4	2D		21	7				
F5	2D		5	7				
F6	2D		6	7				
F7	3D		21	11				
F8	3D		8	11				
F9	3D		9	11				
F10	4D		21	14				
F11	4D		12	14				
F12	4D		13	14				
F13	5D		21	17				
F14	5D		15	17				
F15	5D		16	17				
F16	6D		21	20				
F17	6D		18	20				
F18	6D		19	20				
F19	7D		30	22				
F20	7D		23	22				

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TABLE VII. Output voltage (low-level and high-level) test - Continued.

Test condition		Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
F21	7D	Out voltage, high level	24	22	2.4	4.5	V	9
F22	8D	↓	30	25	↓	↓	↓	↓
F23	8D		26	25				
F24	8D		27	25				
F25	9D		30	28				
F26	9D		29	28				
F27	9D		31	28				
F28	10D		30	32				
F29	10D		33	32				
F30	10D		34	32				
F31	11D		30	35				
F32	11D		36	35				
F33	11D		37	35				
F34	12D		30	38				
F35	12D		39	38				
F36	12D		40	38				

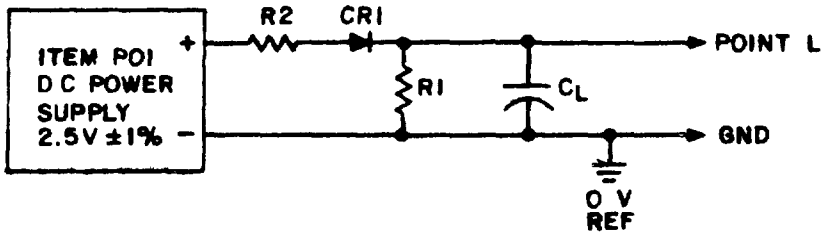
Source: https://assist.dla.mil - Downloaded: 2026-10-20T15:21Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 49.9Ω ±1%, 1 W.
* 2. Set item P01-2 voltage as follows:
V_{IN} = 2.0 V for test conditions E1-E12.
V_{IN} = 0.8 V for test conditions F1-F36.

FIGURE 9. Output voltage (low-level and high-level) test circuit for table VII.

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NOTES:
1. CR1 = 1N4150 or equivalent.
2. C_L = 130 ±5 pF (including probe and parasitic capacitance).
3. R1 = 4.02 kΩ ±1%, 1/8 W.
4. R2 = Select value for a current flow of 22 mA ±1% from point L to ground
5. Capacitor C_L is not required for dc electrical conditions E or F.
6. During output switching, item P01 shall have a maximum overshoot of 500 mV peak.

FIGURE 10. GDA logic load circuit.

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TABLE VIII. Input current (low-level and high-level) and output short circuit current test.

Test condition		Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G1	1B	Input current, low level	3	2	Open	1.0	2.2	mA	11
G2	1C		2	3					
G3	2B		6	5					
G4	2C		5	6					
G5	3B		9	8					
G6	3C		8	9					
G7	4B		13	12					
G8	4C		12	13					
G9	5B		16	15					
G10	5C		15	16					
G11	6B		19	18					
G12	6C		18	19					
G13	7B		24	23					
G14	7C		23	24					
G15	8B		27	26					
G16	8C		26	27					
G17	9B		31	29					
G18	9C		29	31					

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TABLE VIII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G19	10B Input current, low level	34	33	Open	1.0	2.2	mA	11
G20	10C	33	34					
G21	11B	37	36					
G22	11C	36	37					
G23	12B	40	39					
G24	12C	39	40					
G25	1-6A	2,3,5,6,8,9,12,13,15,16,18,19	21		6.0	15.4		
G26	7-12A	23,24,26,27,29,31,33,34,36,37,39,40	30		6.0	15.4		

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TABLE VIII Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H1	1B	Input current, high level	2	Open	21	0	60	µA	11
H2	1B	↓	2	↓	3	↓	↓	↓	↓
H3	1C		3		21				
H4	1C		3		2				
H5	2B		5		21				
H6	2B		5		6				
H7	2C		6		21				
H8	2C		6		5				
H9	3B		8		21				
H10	3B		8		9				
H11	3C		9		21				
H12	3C		9		8				
H13	4B		12		21				
H14	4B		12		13				
H15	4C		13		21				
H16	4C		13		12				
H17	5B		15		21				

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TABLE VIII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H18	5B	Input current, high level	15	Open	16	0	60	µA	11
H19	5C		16		21				
H20	5C		16		15				
H21	6B		18		21				
H22	6B		18		19				
H23	6C		19		21				
H24	6C		19		18				
H25	7B		23		30				
H26	7B		23		24				
H27	7C		24		30				
H28	7C		24		23				
H29	8B		26		30				
H30	8B		26		27				
H31	8C		27		30				
H32	8C		27		26				
H33	9B		29		30				
H34	9B		29		31				

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TABLE VIII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
H35	9C Input current, high level	31	Open	30	0	60	µA	11
H36	9C	31		29				
H37	10B	33		30				
H38	10B	33		34				
H39	10C	34		30				
H40	10C	34		33				
H41	11B	36		30				
H42	11B	36		37				
H43	11C	37		30				
H44	11C	37		36				
H45	12B	39		30				
H46	12B	39		40				
H47	12C	40		30				
H48	12C	40		39				
* H49	1-6A	21		2,3,5,6,8,9,12,13,15,16,18,19	-0.80	-1.15	mA	
* H50	7-12A	30		23,24,26,27,29,31,33,34,36,37,39,40	-0.80	-1.15	mA	

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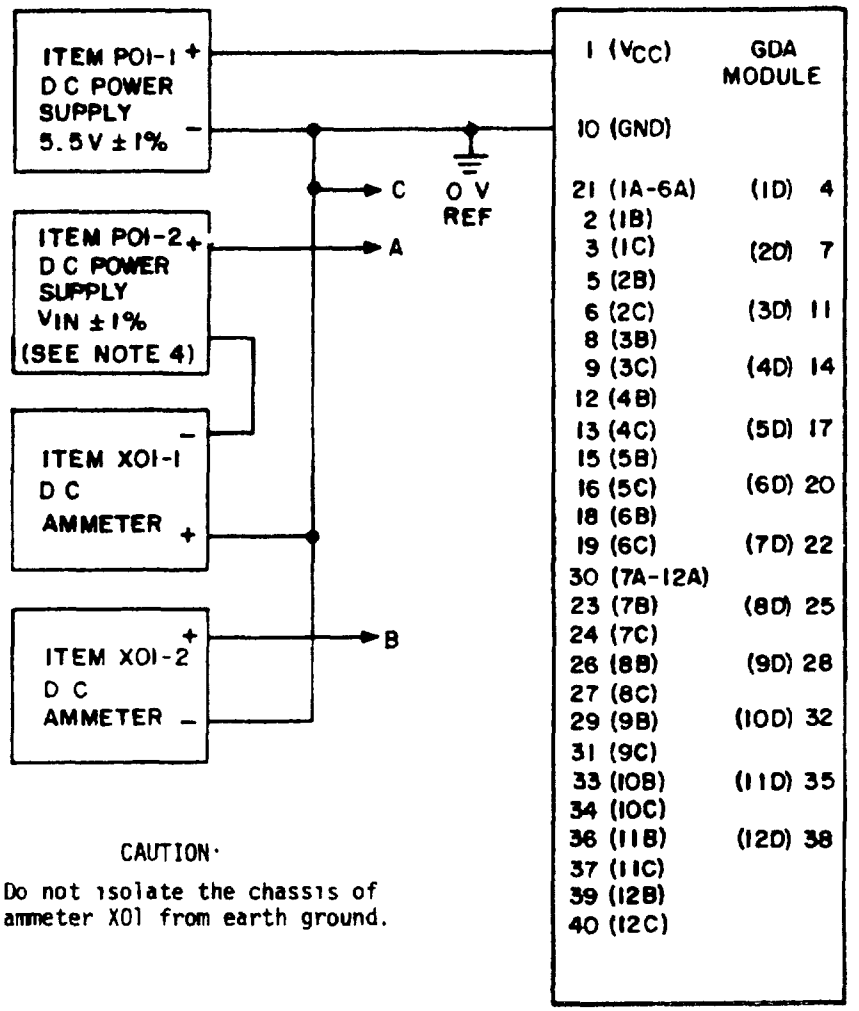
TABLE VIII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
J1	1D	Output short circuit current	Open	4	2,3,21	40	100	mA	11
J2	2D			7	5,6,21				
J3	3D			11	8,9,21				
J4	4D			14	12,13,21				
J5	5D			17	15,16,21				
J6	6D			20	18,19,21				
J7	7D			22	23,24,30				
J8	8D			25	26,27,30				
J9	9D			28	29,31,30				
J10	10D			32	33,34,30				
J11	11D			35	36,37,30				
J12	12D			38	39,40,30				

TABLE IX. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
K1	Power supply current, low level	8	30	120	mA	12
K2	Power supply current, high level	Pin 10	16	52	mA	12

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:

1. Use item X01-2 for test conditions G1-G26 and J1-J12 and item X01-1 for test conditions H1-H50.
2. Since item X01-1 and X01-2 are not used simultaneously, only item X01 is required.
3. With point A disconnected, item X01-1 should indicate 0.0 current on the 1.0 µA range.
* 4. Set item P01-2 voltage as follows:
   V_{IN} = 4.5 V for test conditions G1-G26.
   V_{IN} = 2.4 V for test conditions H1-H50.
5. When performing output short circuit current test, only one output shall be shorted at a time, and the measurement taken within five seconds.

FIGURE 11. Input current (low-level and high-level) and output short circuit current test circuit for table VIII.

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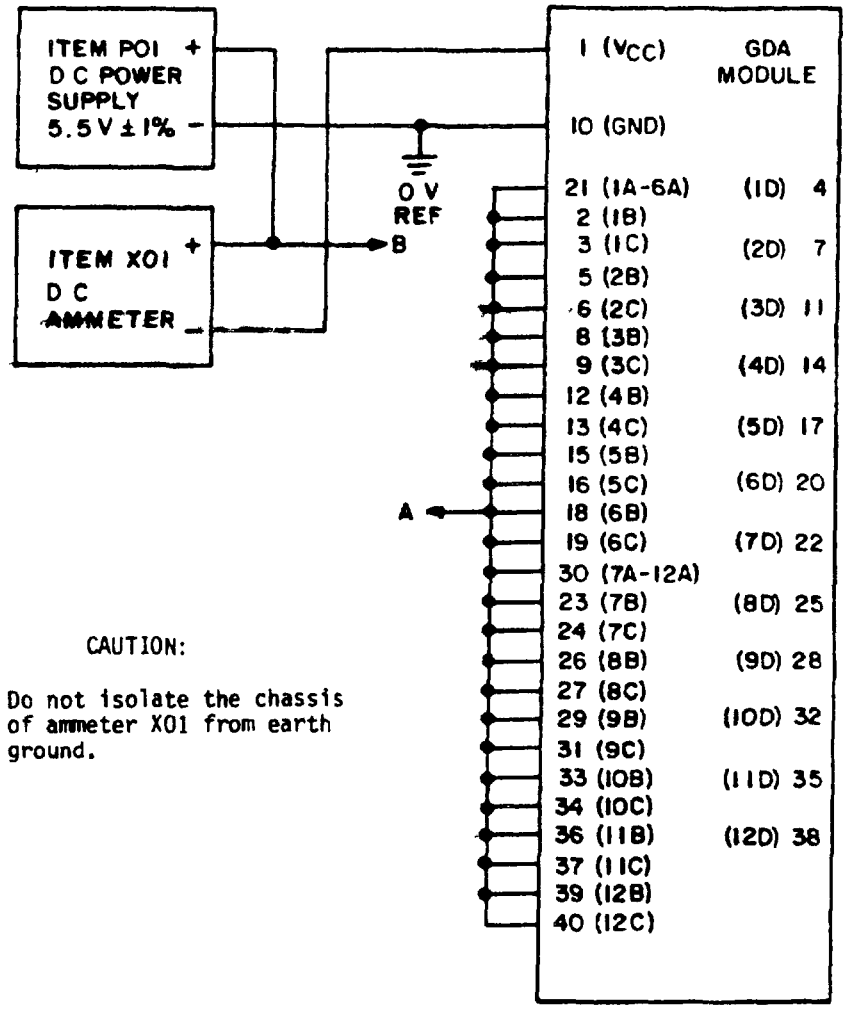


FIGURE 12 Power supply current (low-level and high-level) test circuit for table IX.

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* TABLE X Transition time (t_{THL}) test

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	1B Transition time (t_{THL})	2	4	3	Open	3	20	ns	13,14
L2	2B	5	7	6					
L3	3B	8	11	9					
L4	4B	12	14	13					
L5	5B	15	17	16					
L6	6B	18	20	19					
L7	7B	23	22	Open	24				
L8	8B	26	25		27				
L9	9B	29	28		31				
L10	10B	33	32		34				
L11	11B	36	35		37				
L12	12B	39	38		40				

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* TABLE XI Transition time (t_{TLH}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	1B Transition time (t_{TLH})	2	4	3	Open	3	20	ns	13,14
M2	2B	5	7	6	↓	↓	↓	↓	↓
M3	3B	8	11	9	↓	↓	↓	↓	↓
M4	4B	12	14	13	↓	↓	↓	↓	↓
M5	5B	15	17	16	↓	↓	↓	↓	↓
M6	6B	18	20	19	↓	↓	↓	↓	↓
M7	7B	23	22	Open	24	↓	↓	↓	↓
M8	8B	26	25	↓	27	↓	↓	↓	↓
M9	9B	29	28	↓	31	↓	↓	↓	↓
M10	10B	33	32	↓	34	↓	↓	↓	↓
M11	11B	36	35	↓	37	↓	↓	↓	↓
M12	12B	39	38	↓	40	↓	↓	↓	↓

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:21Z
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* TABLE XII Propagation delay time (t_{PLH}) test.

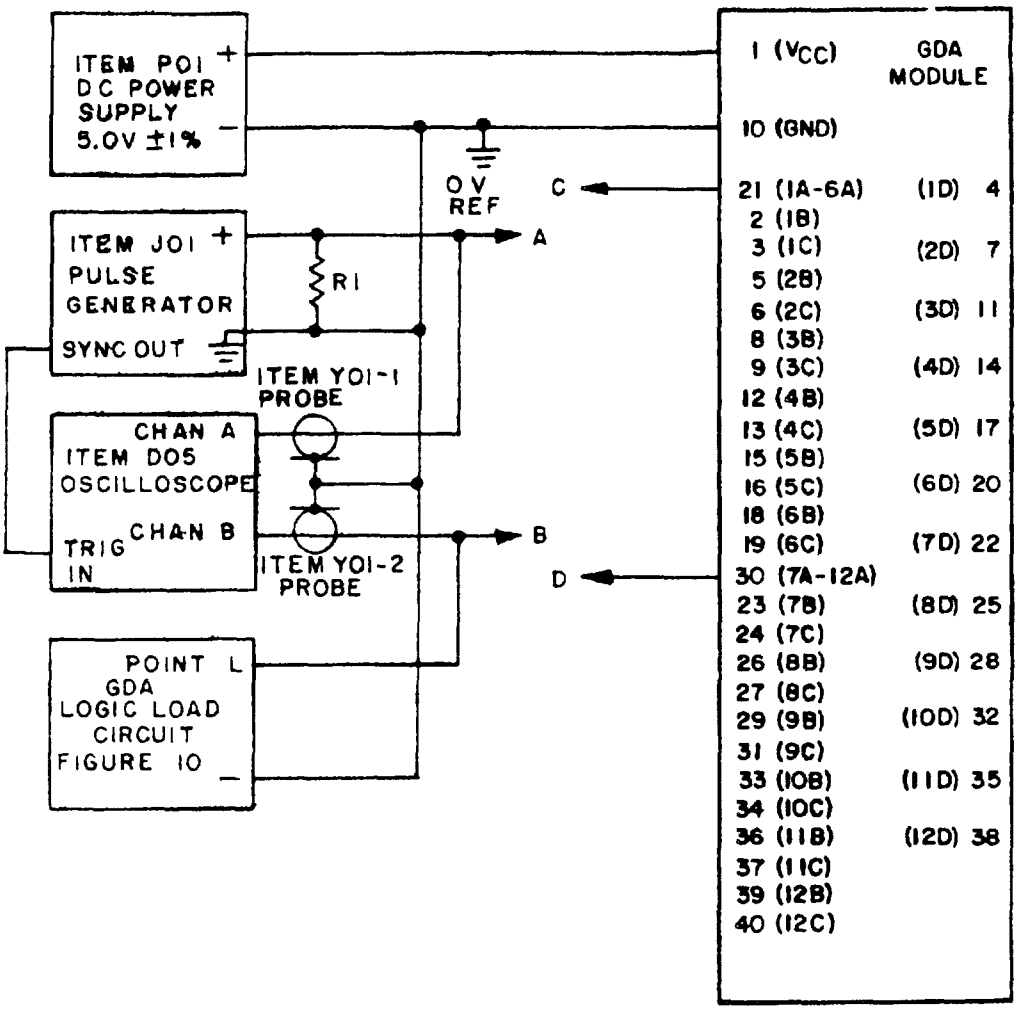
Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	1B	Propagation delay time (t_{PLH})	2	4	3	Open	3	20	ns	13,14
N2	2B		5	7	6					
N3	3B		8	11	9					
N4	4B		12	14	13					
N5	5B		15	17	16					
N6	6B		18	20	19					
N7	7B		23	22	Open	24				
N8	8B		26	25		27				
N9	9B		29	28		31				
N10	10B		33	32		34				
N11	11B		36	35		37				
N12	12B		39	38		40				

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* TABLE XIII. Propagation delay time (t_{PHL}) test.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1B	Propagation delay time (t_{PHL})	2	4	3	Open	3	20	ns	13,14
P2	2B	↓	5	7	6	↓	↓	↓	↓	↓
P3	3B		8	11	9					
P4	4B		12	14	13					
P5	5B		15	17	16					
P6	6B		18	20	19					
P7	7B		23	22	Open	24				
P8	8B		26	25	↓	27				
P9	9B		29	28		31				
P10	10B		33	32		34				
P11	11B		36	35		37				
P12	12B		39	38		40				

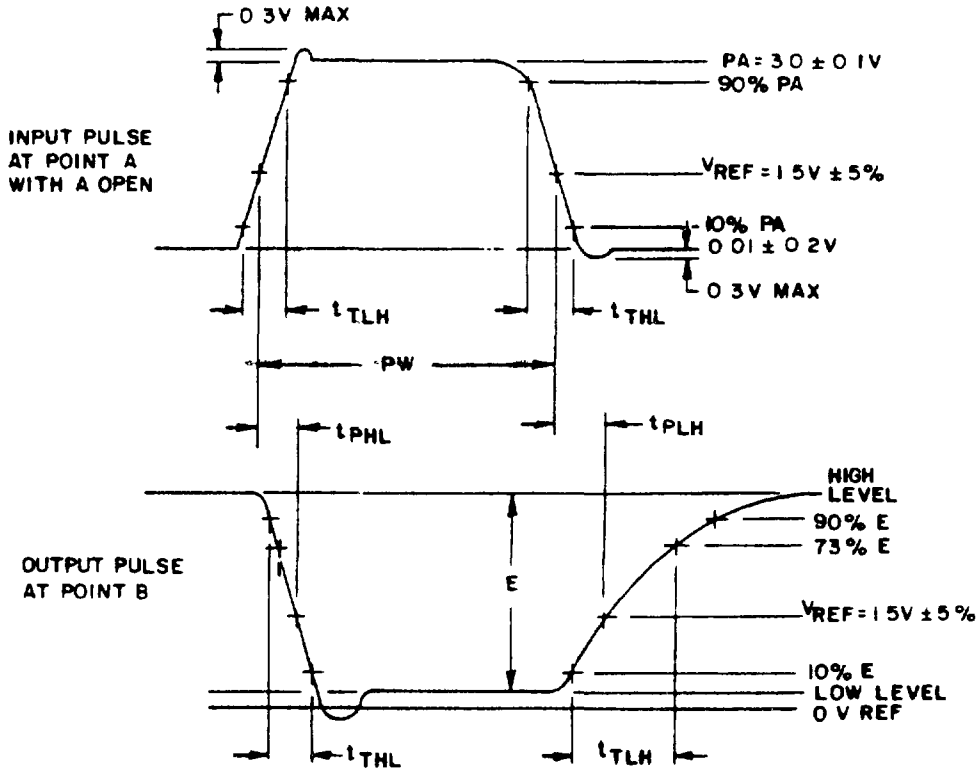
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:21Z
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NOTES
1. R1 = 49.9Ω ±1%, 1 W.
2 Before making measurements, the output levels shall be verified to be 0.45 V maximum for the low level and 2.4 V minimum for the high level

* FIGURE 13. Transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables X, XI, XII and XIII.

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NOTE Input pulse parameters
PW = 500 ns ±10%
$t_{TLH} = t_{THL} = 10 \pm 2$ ns
PRR = 1.0 MHz ±10%

FIGURE 14. Transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables X, XI, XII and XIII.

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TABLE XIV Power supply transient amplitude test

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
J	Power supply transient amplitude	0	400	mV	15, 16

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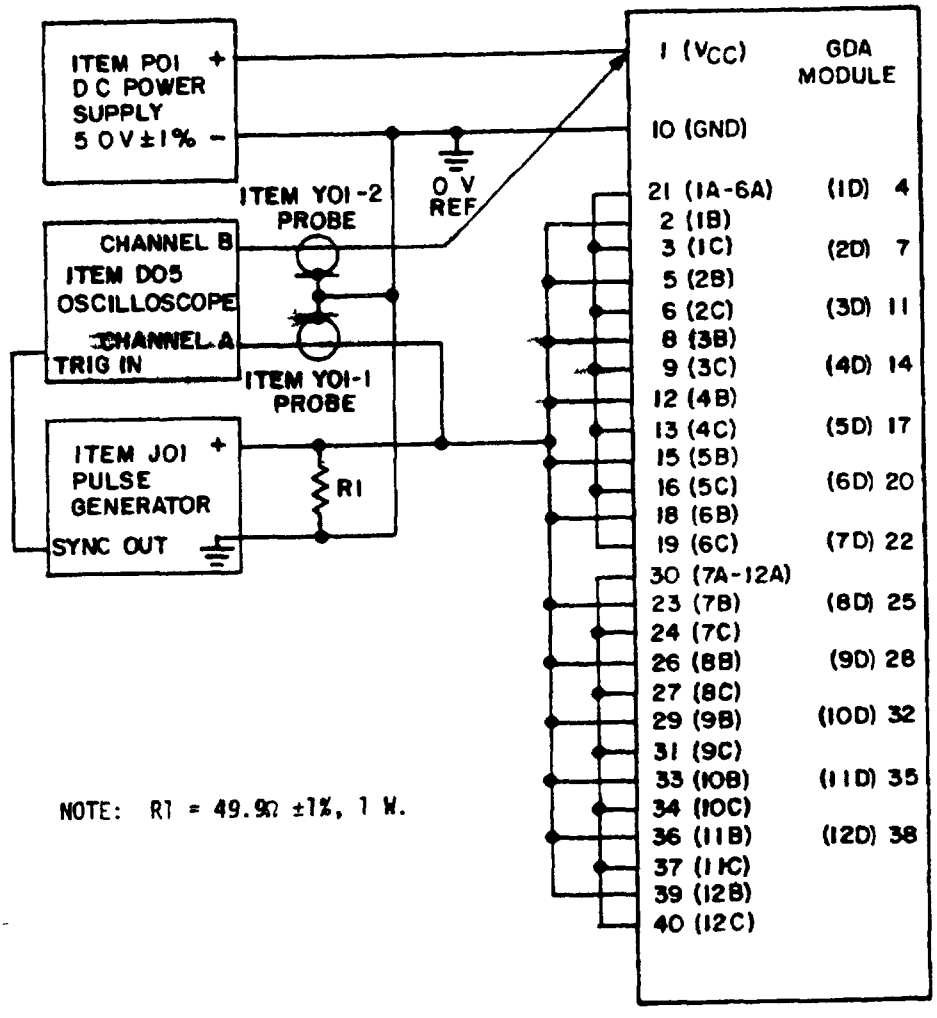
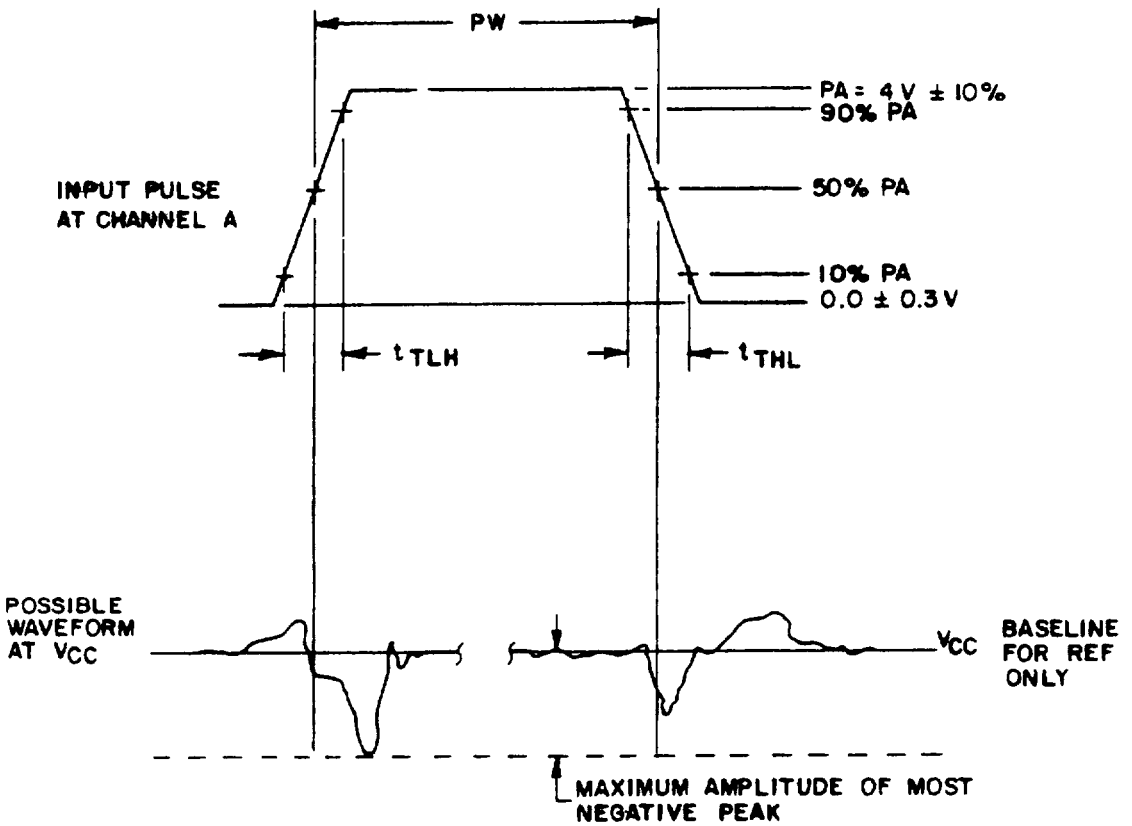


FIGURE 15. Power supply transient amplitude test circuit for table XIV.

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NOTE: Input pulse parameters:
PW = 3 µs ±10%.
$t_{TLH} = t_{THL}$ = 10 ±1 ns.
PRR = 100 kHz ±10%

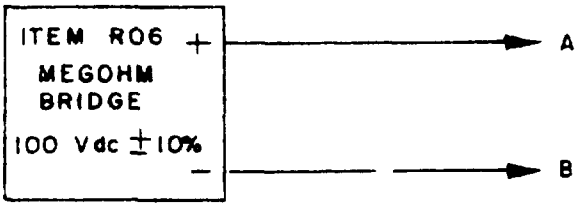
* FIGURE 16. Power supply transient amplitude test circuit waveforms for table XIV.

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* TABLE XV Isolation test

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
R	Isolation	All pins	Frame	10	∞	MΩ	17

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1 (V_{CC})	GDA MODULE
10 (GND)	
21 (1A-6A)	(1D) 4
2 (1B)	
3 (1C)	(2D) 7
5 (2B)	
6 (2C)	(3D) 11
8 (3B)	
9 (3C)	(4D) 14
12 (4B)	
13 (4C)	(5D) 17
15 (5B)	
16 (5C)	(6D) 20
18 (6B)	
19 (6C)	(7D) 22
30 (7A-12A)	
23 (7B)	(8D) 25
24 (7C)	
26 (8B)	(9D) 28
27 (8C)	
29 (9B)	(10D) 32
31 (9C)	
33 (10B)	(11D) 35
34 (10C)	
36 (11B)	(12D) 38
37 (11C)	
39 (12B)	
40 (12C)	

WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge at "discharge" before connecting or disconnecting the test leads.

NOTE Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

* FIGURE 17. Isolation test circuit for table XV.

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* TABLE XVI. Output crosstalk (low-level and high-level) test.

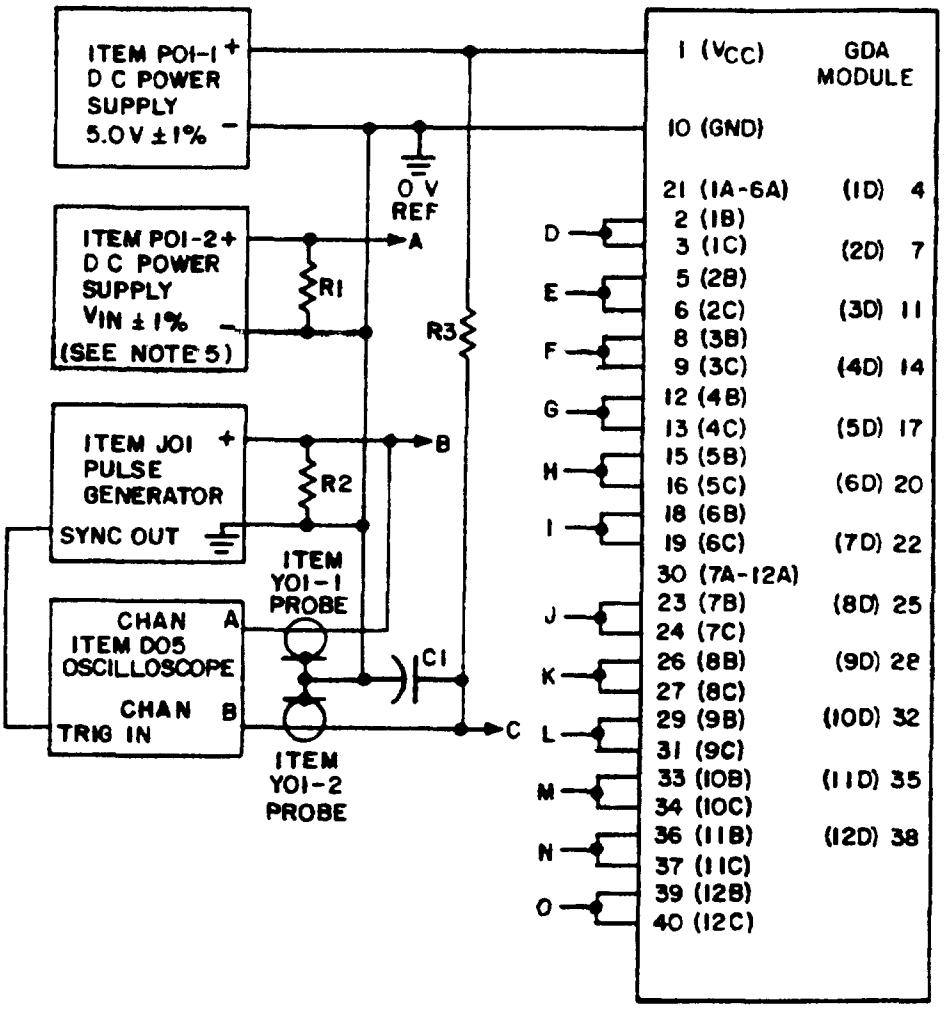
Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
S1	1D Output crosstalk, low level	D	E,F,G,H,I, J,K,L,M,N,O	4	0	400	mV	18, 19
S2	2D	E	D,F,G,H,I, J,K,L,M,N,O	7				
S3	3D	F	D,E,G,H,I, J,K,L,M,N,O	11				
S4	4D	G	D,E,F,H,I, J,K,L,M,N,O	14				
S5	5D	H	D,E,F,G,I, J,K,L,M,N,O	17				
S6	6D	I	D,E,F,G,H, J,K,L,M,N,O	20				
S7	7D	J	D,E,F,G,H, I,K,L,M,N,O	22				
S8	8D	K	D,E,F,G,H, I,J,L,M,N,O	25				
S9	9D	L	D,E,F,G,H, I,J,K,M,N,O	28				
S10	10D	M	D,E,F,G,H, I,J,K,L,N,O	32				
S11	11D	N	D,E,F,G,H, I,J,K,L,M,O	35				
S12	12D	O	D,E,F,G,H, I,J,K,L,M,N	38				

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:21Z
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* TABLE XVI. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
T1	1D	Output crosstalk, high level	D	E,F,G,H,I, J,K,L,M,N,O	4	0	400	mV	18, 19
T2	2D		E	D,F,G,H,I, J,K,L,M,N,O	7				
T3	3D		F	D,E,G,H,I, J,K,L,M,N,O	11				
T4	4D		G	D,E,F,H,I, J,K,L,M,N,O	14				
T5	5D		H	D,E,F,G,I, J,K,L,M,N,O	17				
T6	6D		I	D,E,F,G,H, J,K,L,M,N,O	20				
T7	7D		J	D,E,F,G,H, I,K,L,M,N,O	22				
T8	8D		K	D,E,F,G,H, I,J,L,M,N,O	25				
T9	9D		L	D,E,F,G,H, I,J,K,M,N,O	28				
T10	10D		M	D,E,F,G,H, I,J,K,L,N,O	32				
T11	11D		N	D,E,F,G,H, I,J,K,L,M,O	35				
T12	12D		O	D,E,F,G,H, I,J,K,L,M,N	38				

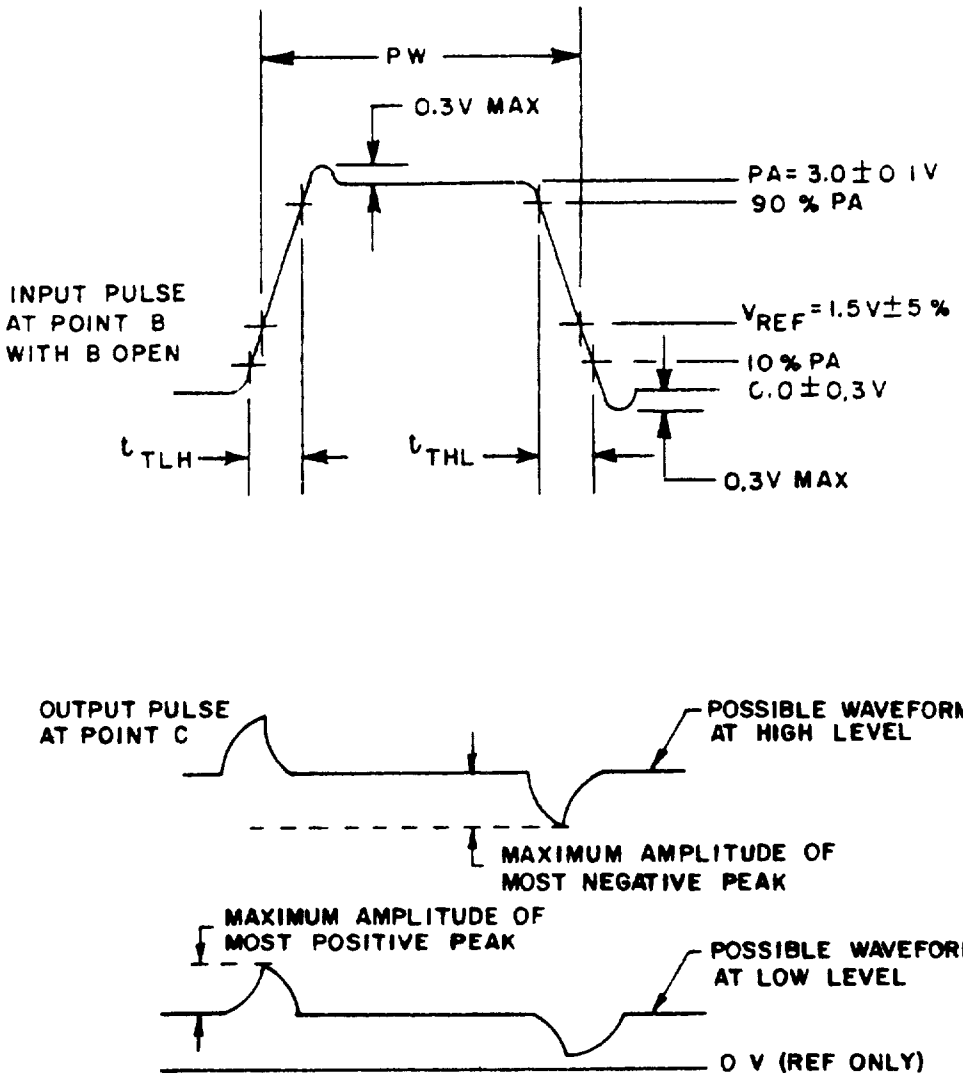
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NOTES
1. R1 = R2 = 49.9Ω ±1%, 1 W.
2. R3 = 243Ω ±1%, 1/4 W.
3. C1 = 30 pF ±1%.
4. Only the output under test shall be loaded. The loop length from pin 1 to each load resistor and output, including the length of the resistor body, shall not exceed 4.5 inches (114.3 mm).
* 5. Set item PO1-2 voltage as follows:
V_{IN} = 2.4 V for test conditions S1-S12.
V_{IN} = 400 mV for test conditions T1-T12.

FIGURE 18. Output crosstalk (low-level and high-level) test circuit for table XVI.

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NOTE Input pulse parameters:
PW = 200 ns ±10%
$t_{TLH} = t_{THL}$ = 25 ±5 ns.
PRR = 1.0 MHz ±10%

* FIGURE 19. Output crosstalk (low-level and high-level) test circuit waveforms for table XVI.

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TABLE XVII. Input capacitance test.

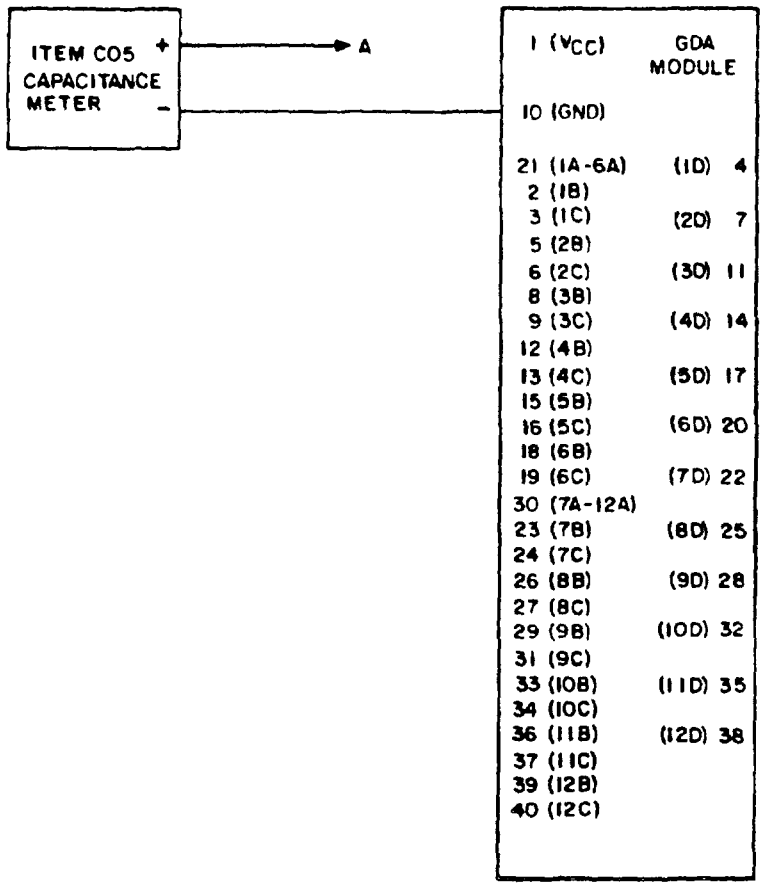
Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
U1	1B	Input capacitance	2	2	12	pF	20
U2	1C		3				
U3	2B		5				
U4	2C		6				
U5	3B		8				
U6	3C		9				
U7	4B		12				
U8	4C		13				
U9	5B		15				
U10	5C		16				
U11	6B		18				
U12	6C		19				
U13	7B		23				
U14	7C		24				
U15	8B		26				
U16	8C		27				
U17	9B		29				

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TABLE XVII Input capacitance test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
U18	9C	Input capacitance	31	2	12	pF	20
U19	10B	↓	33	↓	↓	↓	↓
U20	10C		34				
U21	11B		36				
U22	11C		37				
U23	12B		39				
U24	12C		40				
U25	1A-6A		21	15	60		
U26	7A-12A		30	15	60		

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NOTES
1 Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings
2 Null out the effects of the test leads prior to connecting to the module

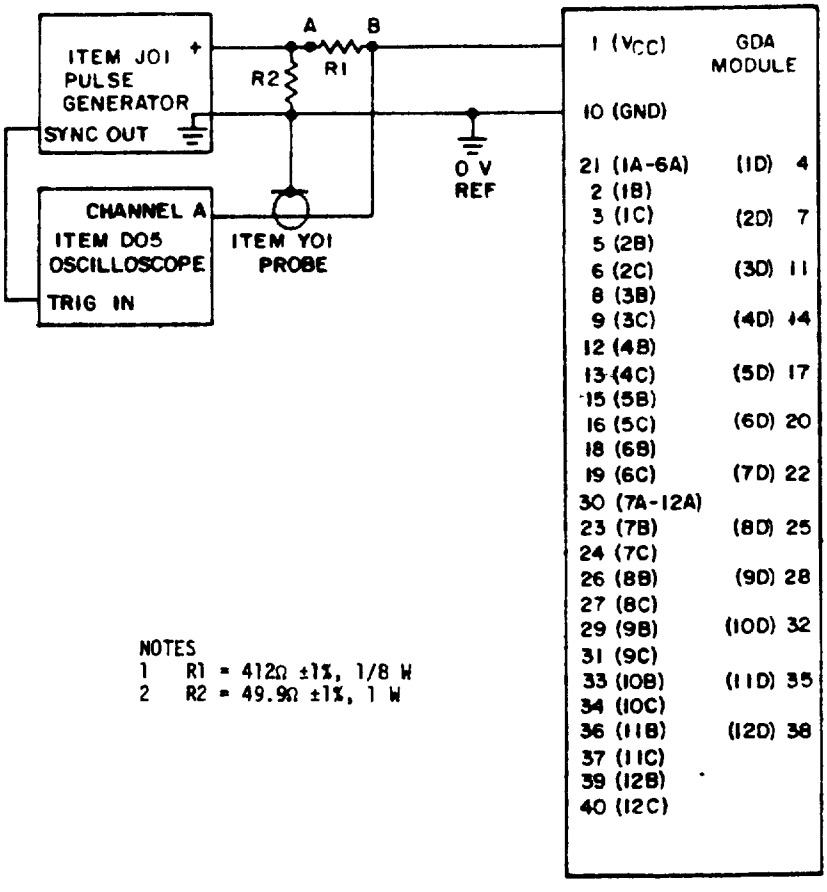
FIGURE 20 Input capacitance test circuit for table XVII

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TABLE XVIII Power supply transient rise time test

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
V	Power supply transient rise time	2.5	7.5	µs	21,22

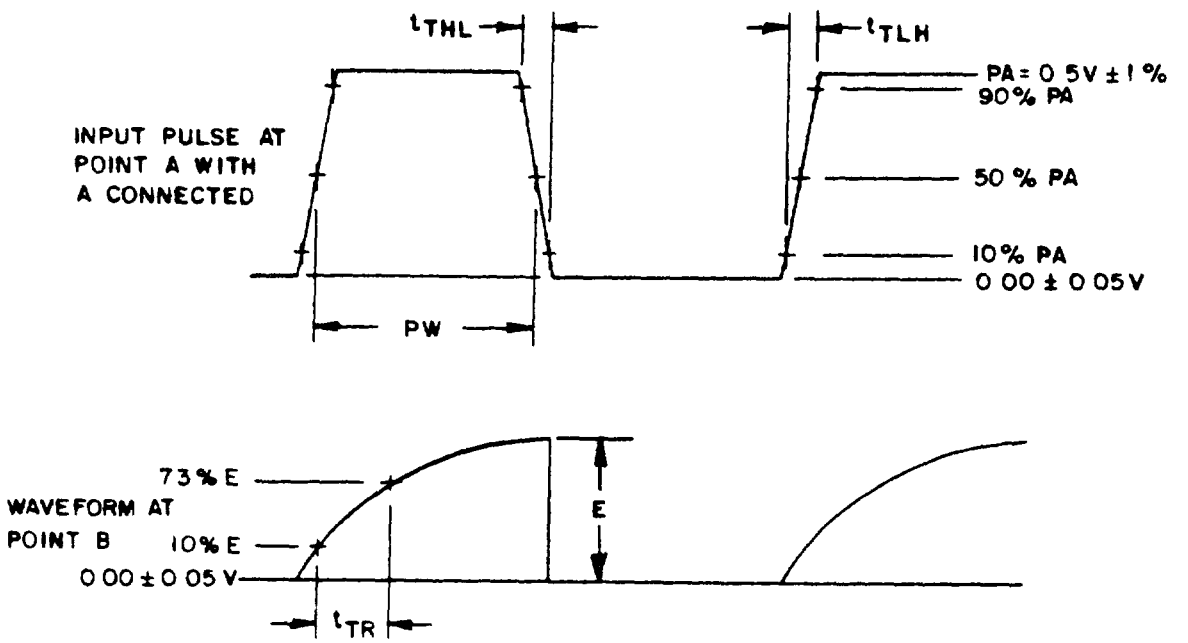
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NOTES
1 R1 = 412Ω ±1%, 1/8 W
2 R2 = 49.9Ω ±1%, 1 W

FIGURE 21. Power supply transient rise time test circuit for table XVIII.

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NOTES:
1. Input pulse parameters:
   PW = 30 µs ±10%.
   $t_{TLH} = t_{THL}$ = 50 ns ±10%.
   PRP = 60 µs ±10%.
2. Measure transient rise time (t_{TR}) at point B.

* FIGURE 22. Power supply transient rise time test circuit waveforms for table XVIII.

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INCH-POUND

MIL-M-28787/3B(SH)
AMENDMENT 2
29 March 1991
SUPERSEDING
AMENDMENT 1
26 April 1985

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE GDA

This amendment forms a part of Military Specification MIL-M-28787/3B(EC), dated 3 June 1982, and is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

PAGE 4

* TABLE I, Test conditions D1-D24, Min columns. Delete "-0 4" (four places) and substitute "-1.5" (four places).

* TABLE I, Test conditions D1-D24, Max columns Delete "-1.5" (four places) and substitute "-0.4" (four places).

* TABLE I, Test conditions D25, D26, Initial acceptance requirements, Min column Delete "-0 4" (one place) and substitute "-1.5" (one place)

* TABLE I, Test conditions D25, D26, Initial acceptance requirements, Max column Delete "-1.5" (one place) and substitute "-0.4" (one place)

* TABLE I, Test conditions D25, D26, End-of-life acceptance requirements, Min column Delete "-0.4" (one place) and substitute "-1.51" (one place).

* TABLE I, Test conditions D25, D26, End-of-life acceptance requirements, Max column Delete "-1.51" (one place) and substitute "-0.4" (one place)

* TABLE I, Test conditions D25, D26, Operating temperature requirements, Min columns Delete "-0 4" (two places) and substitute "-1.5" (two places).

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* TABLE I, Test conditions D25, D26, Operating temperature requirements, Max columns Delete "-1 5" (two places) and substitute "-0 4" (two places)

* TABLE I, Test conditions H49, H50, Min columns Delete "-0 80" (four places) and substitute "-1 66" (four places)

* TABLE I, Test conditions H49, H50, Max columns Delete "-1.15" (four places) and substitute "-0 80" (four places)

TABLE I, test conditions J1-J12, Test type column Delete "X" and substitute "S"

PAGE 6

* 4 4.2 Delete and substitute the following

"4 4.2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. Test conditions R, A, B, D, and J are stress tests and shall be performed only at 25 ± 5°C. If test conditions R, A, B, D, or J are part of a sequence of tests, the order of testing shall be as follows. R, A, B, D, and J where applicable, followed by the remaining tests in the sequence "

* 4 5 1 Delete third sentence and substitute the following

"When electrical testing is called for at the end of each temperature stabilization period, test conditions C, E through H, K through Q, S, T, and V shall be performed in the order specified in 4 4 2 using the initial limits for applicable operating temperature "

* 4 5 3 Delete and substitute the following.

"4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787 The module shall be operated as shown on figure 2 Following the life test and while still at the maximum class temperature, electrical tests C, E through H, K through Q, S, T, and V shall be performed in the order specified in 4 4.2 using the operating temperature end-of-life requirements in table I Following the return of the module to 25 ± 5°C, electrical tests C, E through H, K through Q, S, T, and V shall be performed in the order specified in 4 4 2 using the 25 ± 5°C ambient end-of-life requirements in table I "

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